

Cambridge O Level Chemistry 5070

Ion tests and gas tests

Question paper extracts + mark schemes

Papers 1 not used. Papers 2 and 3 only.

2015 – 2025

Paper 3: written parts only

(describe the test / identify the ion or gas

— the same skill as Paper 2)

Colours inverted.

Types covered:

Anions Cl^- Br^- I^- SO_4^{2-} NO_3^- CO_3^{2-} / CO_2 SO_3^{2-} / SO_2

Cations NH_4^+ Al^{3+} Zn^{2+} Cu^{2+} Fe^{2+} Fe^{3+} Cr^{3+}

Gases NH_3 CO_2 Cl_2 H_2 O_2 SO_2

Flame Na^+ yellow K^+ lilac Cu^{2+} blue-green

Contents

2015-2025 · inverted extracts

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- 02 Paper 2 Chloride ions Cl^- (nitric acid then AgNO_3)
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- 06 Paper 2 Sulfate ions SO_4^{2-}
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- 10 Paper 2 Ammonium ions NH_4^+
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- 13 Paper 2 Carbon dioxide CO_2
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- 43 Paper 3 Flame test Cu^{2+} blue-green; CO_2 from carbonate
- 44 Paper 3 CO_2 ; $\text{Zn}^{2+}/\text{Al}^{3+}$ with NaOH/ NH_3 ; identify ions

(iii) Describe a test for chloride ions.

test

observation

[2]

[Total: 13]

5070/21

Cambridge O Level – Mark Scheme
PUBLISHED

October/November 2019

Question	Answer	Marks
2(e)(ii)	<i>anode:</i> chlorine AND <i>cathode:</i> hydrogen	1
2(e)(iii)	(add nitric acid) then (aqueous) silver nitrate (1) white precipitate (1)	2

Question	Answer	Marks						
3(a)	Any three from: (property on which distillation) depends is the boiling point / copper(II) sulfate has higher boiling point than water / ORA (1) - idea of distillation apparatus, e.g. flask connected to condenser (1) - flask or solution heated (1) - idea that only water vaporised (when flask heated) (1) - water vapour converted to (liquid) water (in condenser) (1)	3						
3(b)	filtration	1						
3(c)	<table border="0"> <tr> <td>Cu</td> <td>Cs</td> <td>Cl</td> </tr> <tr> <td>$\frac{21.09}{64}$</td> <td>$\frac{43.82}{133}$</td> <td>$\frac{35.09}{35.5}$</td> </tr> </table> OR Cu = 0.33 Cs = 0.33 Cl = 0.99 (1) CuCsCl_3 (1)	Cu	Cs	Cl	$\frac{21.09}{64}$	$\frac{43.82}{133}$	$\frac{35.09}{35.5}$	2
Cu	Cs	Cl						
$\frac{21.09}{64}$	$\frac{43.82}{133}$	$\frac{35.09}{35.5}$						

Describe a test for chloride ions.

test

result[2]

(d) Zinc oxide reacts with both hydrochloric acid and sodium hydroxide.

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- (c) (i) correct M_r of 128 **or** (2×64) as numerator of fraction [2]
OR
correct M_r $(2 \times 64) + 12 + (16 \times 5) + (2 \times 1)$ **or** 222 as denominator (1)
percentage = $57.65/57.7$ (1)

- (ii) add acid (1) [2]
gas evolved turns limewater milky (1)

- (d) **A** is oxidation because electrons are lost (1) [2]

B is reduction because electrons are gained (1)

[Total: 10]

- A5 (a) (i) ANY FOUR FROM:** [4]

ammonia molecules/HBr molecules have enough energy to escape from the
 HBr(aq) or $\text{NH}_3(\text{aq})$ (1)

diffusion (1)

molecules move randomly/molecules spread out/molecules get mixed up
(1)

move from high to low concentration/move with the concentration gradient
(1)

solid formed where NH_3 and HBr react (1)

HBr has higher M_r than NH_3 /molecules of HBr are heavier than molecules of
 NH_3 (1)

NH_3 molecules move faster than HBr molecules/ NH_3 diffuses faster (1)

- (b) higher pressure pushes molecules closer together [1]

[Total: 5]

- A6 (a)** mol of $\text{NaOH} = 0.30$ (1) [2]

energy released $(= 0.30 \times 57.1) = 17/17.1(3)$ (kJ) (1)

- (b) mol of $\text{HCl} = 2.19/36.5$ **OR** $= 0.06$ (1) [2]
volume $= (0.06/0.2) = 0.3 \text{ dm}^3/300 \text{ cm}^3$ (1)

- (c) add nitric acid and silver nitrate (1) [2]
white precipitate/white solid formed (1)

- (c) Describe a test for aqueous bromide ions. Include the observations for a positive result.

test

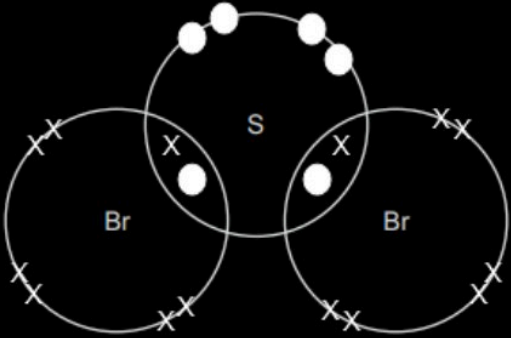
observations

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Cambridge O Level – Mark Scheme
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Question	Answer	Marks
3(d)	the particles move further apart	1

Question	Answer	Marks
4(a)	anode: bromine (1) cathode: hydrogen (1)	2
4(b)	anode: $2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{e}^-$ (1) cathode: $\text{Ca}^{2+} + 2\text{e}^- \rightarrow \text{Ca}$ (1)	2
4(c)	add (acidified) silver nitrate (1) cream precipitate (1)	2
4(d)(i)	calcium gives electrons to zinc <u>ions</u> / electrons transferred from calcium to zinc <u>ions</u>	1
4(d)(ii)	+2	1
4(e)	 pair of electrons between each S and Br (1) 6 non-bonding e⁻ on each Br and 4 non-bonding e⁻ on S (1)	2

(c) Describe a chemical test for the iodide ion.

test

observation

.....

5070/22

Cambridge O Level – Mark Scheme
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May/June 2019

Question	Answer	Marks
7(a)	ammonium iodide disappears / solid disappears / sample disappears / nothing left in tube (1)	1
7(b)	moles of ammonium iodide $\frac{2.9}{145}$ OR 0.02(00) (1) moles of gas = 2×0.02 OR 0.04(0) (1) volume of gas = $(0.04 \times 24) = 0.96 \text{ dm}^3$ OR 960 cm ³ (1)	3
7(c)	(acidified aqueous) silver nitrate (1) (pale) yellow ppt (1)	2
7(d)	$2\text{I}^-(\text{aq}) + \text{Br}_2(\text{aq}) \rightarrow \text{I}_2(\text{aq}) + 2\text{Br}^-(\text{aq})$ correct formulae and balanced (1) correct state symbols – dependent on correct formulae (1)	2
7(e)	in solid ions cannot move (1) in aqueous / solution ions can move (1)	2

- (b) Acidified aqueous silver nitrate is added to aqueous zinc iodide.
State the observations made.

..... [1]

5070/22

Cambridge O Level – Mark Scheme
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October/November 2021

Question	Answer	Marks				
4(d)	C₂H₃ (2) if 2 marks not scored 1 mark for: <table><tr><td>C</td><td>H</td></tr><tr><td>$\frac{88.9}{12}$</td><td>$\frac{11.1}{1}$</td></tr></table> OR 7.4 11.1 (1)	C	H	$\frac{88.9}{12}$	$\frac{11.1}{1}$	2
C	H					
$\frac{88.9}{12}$	$\frac{11.1}{1}$					

Question	Answer	Marks
5(a)	lithium oxidised because it loses electrons (1) zinc ions reduced because they gain electrons (1)	2
5(b)	yellow precipitate / yellow solid	1
5(c)(i)	iodine AND zinc bromide	1
5(c)(ii)	chlorine is more reactive than bromine / bromine is less reactive than chlorine	1

Question	Answer	Marks
6(a)	reactants on the left and products on the right and reactant line below product line (1) arrow upwards between reactants and products with ΔH label (1)	2
6(b)	bond breaking endothermic and bond making exothermic / (thermal) energy absorbed to break bonds and (thermal) energy released on making bonds (1) more energy absorbed than released (1)	2
6(c)	$4\text{PH}_3 + 8\text{O}_2 \rightarrow \text{P}_4\text{O}_{10} + 6\text{H}_2\text{O}$	1

(b) Describe a test for sulfate ions.

test

result

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Cambridge O Level – Mark Scheme
PUBLISHEDOctober/November
2017

Question	Answer	Marks
B7(a)(i)	Magnesium has strong bonding between positive ions / cations and electrons / magnesium is a giant structure (1) Sulfur is a simple molecule / weak forces between sulfur molecules (1)	2
B7(a)(ii)	Magnesium has electrons which move (from place to place) (1) Sulfur does not have delocalised electrons / no mobile electrons / electrons don't move (1)	2
B7(b)	Giant structure / many covalent <u>bonds</u> (1) Need high temperature / lot of energy to break the <u>bonds</u> (1)	2
B7(c)(i)	Mass of sulfur = 19.2 g (1) $\text{mol S} = \frac{19.2}{32}$ $\text{mol Cl} = \frac{21.3}{35.5}$ OR ratio = 0.6 to 0.6 (1) SCl (1)	3
B7(c)(ii)	S_2Cl_2 (1)	1

Question	Answer	Marks
B8(a)	$\frac{2 \times 39}{174} \times 100 = 44.8\% / 45\%$ (2 marks) If 2 marks not scored correct $M_r = 174$ (1)	2
B8(b)	(Acidified) barium chloride / barium nitrate (1) White precipitate (1)	2
B8(c)	Nitrates soluble (in water) / nitrates dissolve (easily) (1)	1

(d) Describe the observations when aqueous barium nitrate is added to aqueous copper(II) sulfate.

..... [1]

(e) Aqueous copper(II) sulfate is electrolysed separately with carbon electrodes and with

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Question	Answer	Marks
8(a)	$\text{C}_{14}\text{H}_{30}$	1
8(b)	tetradecane	1
8(c)(i)	M1 fractional distillation / use of a fractionating column (1) M2 petroleum is heated / petroleum is vapourised / petroleum is boiled (1) M3 separation because of different boiling points (1)	3
8(c)(ii)	heating / cooking	1
8(c)(iii)	(chemical) feedstock	1

Question	Answer	Marks									
9(a)	substance which contains no water	1									
9(b)	blue	1									
9(c)	(pale) blue precipitate (1) soluble in excess resulting in dark blue solution (1)	2									
9(d)	white precipitate	1									
9(e)	<table border="1"> <tr> <td>electrodes</td><td>carbon</td><td>copper</td></tr> <tr> <td>anode</td><td>fizzing / bubbling (1)</td><td>gets smaller (1)</td></tr> <tr> <td>cathode</td><td>pink solid (1)</td><td>gets bigger / pink solid (1)</td></tr> </table>	electrodes	carbon	copper	anode	fizzing / bubbling (1)	gets smaller (1)	cathode	pink solid (1)	gets bigger / pink solid (1)	4
electrodes	carbon	copper									
anode	fizzing / bubbling (1)	gets smaller (1)									
cathode	pink solid (1)	gets bigger / pink solid (1)									
9(f)	ions can move in aqueous / ions cannot move in solid	1									

- (c) reacts with warm aqueous sodium hydroxide and aluminium foil to give a gas that turns damp red litmus paper blue

..... [1]

- (d) has an aqueous solution that is used to test for an oxidising agent

..... [1]

- (e) has an aqueous solution that reacts with copper metal.

..... [1]

[Total: 5]



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Question	Answer	Marks
1(a)	barium sulfate	1
1(b)	copper(II) sulfate	1
1(c)	silver nitrate	1
1(d)	potassium iodide	1
1(e)	silver nitrate	1

Question	Answer	Marks
2(a)(i)	each carbon atom in the structure uses three outer shell electrons in forming covalent bonds (1) presence of mobile electrons (between layers) (1)	2
2(a)(ii)	inert	1
2(b)	(anode) bromine / Br_2 (1) (cathode) copper / Cu (1)	2
2(c)	$4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$ / $4\text{OH}^- - 4\text{e}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O}$ (1) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ (1)	2
2(d)(i)	water	1
2(d)(ii)	idea of difficult to store the hydrogen under pressure / availability of hydrogen sources	1

(c) Describe a chemical test for the ammonium ion.

test

observation

..... [2]

(d) Aqueous ammonium carbonate reacts with dilute hydrochloric acid.

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Question	Answer	Marks
7(a)	solid disappears / ammonium carbonate disappears / nothing left in tube (1)	1
7(b)	moles of ammonium carbonate = $4.80 / 96$ OR $0.05(00)$ (1) moles of gas = 3×0.05 OR 0.15 (1) volume of gas = $(0.15 \times 24) = 3.6 \text{ dm}^3$ OR 3600 cm^3 (1)	3
7(c)	heat or warm with (aqueous) sodium hydroxide (1) gas that turns (moist red) litmus blue (1)	2
7(d)	$\text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ correct formulae (1) correct state symbols – dependent on correct formulae (1)	2
7(e)	in solid ions cannot move (1) in aqueous solution ions can move (1)	2

Question	Answer	Marks
8(a)	when a reversible reaction (in a closed system) (1) (reaches a point that the) rate of forward reaction equals the rate of the backward reaction (1)	2
8(b)	more PCl_5 / concentration of PCl_5 increases / less PCl_3 / less Cl_2 / concentration of Cl_2 decreases / concentration of PCl_3 decrease (1) there are fewer moles of gas on the left hand side of the reaction (1)	2
8(c)(i)	the reaction absorbs heat / the (forward) reaction is endothermic (1)	1

(b) Describe a test for ammonium ions.

test

observation

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Page 4	Mark Scheme	Syllabus	Paper
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Question	Answer	Marks
A3(d)(ii)	Any two of: carbon monoxide/carbon/water (1) carbon monoxide is toxic/carbon monoxide is poisonous (1)	2
	Total:	11

Question	Answer	Marks
A4(a)	propane: increases (rate)/doubling concentration doubles rate (1) iodine: no effect (on rate) (1) hydrochloric acid: increases (rate)/doubling concentration doubles rate (1)	3
A4(b)	particles move faster/particles have more energy (1) more particles have (energy greater than) the activation energy/more successful collisions (1)	2
A4(c)	<u>atom(s)</u> with same <u>number</u> of protons and different number of neutrons/ <u>atom(s)</u> of the same element with different <u>number</u> of neutrons (1)	1
A4(d)	$I_2 + 2At^- \rightarrow At_2 + 2I^-$ (1)	1
A4(e)	they lose electrons/they are oxidised/they give electrons (to hydrogen peroxide)/the oxidation number of iodine increases/the oxidation number of oxygen is decreased (1)	1
	Total:	8

(c) Describe a test for ammonia.

test

result

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2017

Question	Answer	Marks
B8(b)	to increase plant growth / to help make more protein / to add nitrogen to soil depleted by previous year's growth (1)	1
B8(c)	(damp) red litmus paper (1) turns blue (1)	2
B8(d)	ammonia is formed (1) idea of ammonia escaping from the soil as a gas (1)	2
B8(e)	$\text{mol H}_2\text{SO}_4 = 0.150 \times \frac{10.5}{1000}$ OR 1.575×10^{-3} (1) $\text{mol NH}_3(\text{aq}) = 2 \times 1.575 \times 10^{-3}$ OR 3.15×10^{-3} (1) concentration of $\text{NH}_3(\text{aq}) = 0.158 \text{ (mol dm}^{-3}\text{)}$ (1)	3

Question	Answer	Marks
B9(a)	they have the same molecular formula but the atoms are arranged differently (1)	1
B9(b)	rate decreases (1) the gradient of the graph decreases / slope of the graph decreases (1)	2
B9(c)	increases rate (no marks) particles closer together / more particles in a given volume / more crowded particles (1) greater collision frequency / more collisions per second / rate of collisions increases (1)	2

(b) Describe a chemical test for carbon dioxide.

test

observation if carbon dioxide present

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Question	Answer	Marks
1(a)(i)	A	1
1(a)(ii)	B	1
1(a)(iii)	F	1
1(a)(iv)	C	1
1(a)(v)	E	1
1(b)	C and D	1

Question	Answer	Marks
2(a)	different number of neutrons / different mass number / different nucleon number	1
2(b)	same electronic configuration	1
2(c)	$\frac{(95.04 \times 32) + (0.75 \times 33) + (4.20 \times 34) + (0.01 \times 36)}{100}$ correct numerator (1) correct denominator (1)	2

Question	Answer	Marks
3(a)(i)	A activation energy / E_a (1) B enthalpy change / ΔH (1)	2
3(a)(ii)	product level above reactant level	1
3(b)	limewater goes milky	1

(b) Describe a test for carbon dioxide.

test

observation

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Question	Answer	Marks
1(a)	cobalt(II) chloride	1
1(b)	iron(III) chloride	1
1(c)	silver chloride	1
1(d)	ammonium chloride	1
1(e)	hydrogen chloride	1

Question	Answer	Marks
2(a)	carbon monoxide (1) octane (1)	2
2(b)	limewater (1) turns milky / turns cloudy / white precipitate (1)	2
2(c)	decomposition of vegetation / decomposition of plants	1
2(d)	greenhouse (1) global warming (1)	2
2(e)	mention of nitrogen oxides and carbon monoxide (as the pollutants / reactants) (1) nitrogen / N ₂ (in exhaust gases) (1) carbon dioxide / CO ₂ (in exhaust gases) (1)	3

- (i) Name **Y** and describe a test for this gas.

name

test

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Question	Answer	Marks
8(d)(i)	carbon dioxide (1) limewater turns milky (1)	2
8(d)(ii)	water (1) copper(II) sulfate goes from white to blue / cobalt chloride goes from blue to pink (1)	2
8(d)(iii)	$\text{CaCO}_3 + 2\text{HNO}_3 \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{CO}_2 + \text{H}_2\text{O}$ (1)	1

Question	Answer	Marks
9(a)	SO_3NH_3 (1)	1
9(b)(i)	(a substance that) donates hydrogen ions / (a substance that) produces hydrogen ions (in solution) (1)	1
9(b)(ii)	weak acids partially ionise / weak acids do not completely dissociate / weak acids do not fully ionise (1) strong acids completely ionise / strong acids completely dissociate (1)	2
9(c)	moles of acid = $0.25 \times 0.15 / 0.0375$ (1) M_r of acid = 97 (1) mass of acid (= moles of acid $\times M_r$) = 3.6(4) (1)	3
9(d)	16.7 (cm ³) (1)	1
9(e)	$\text{Mg} + 2\text{SO}_3\text{NH}_3 \rightarrow \text{H}_2 + \text{Mg}(\text{SO}_3\text{NH}_2)_2$ correct formula of $\text{Mg}(\text{SO}_3\text{NH}_2)_2$ (1) balanced equation – dependent on correct formulae (1)	2

(ii) Describe the chemical test for sulfur dioxide.

test

.....

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Question	Answer	Marks
B10(a)(i)	Moles of Zn and H ₂ = 0.01 (1) Volume of H ₂ = 0.24 dm ³ (1)	2
B10(a)(ii)	The number of moles of iron is different (1)	1
B10(b)	One mark each for any two observations from: Solution gets warmer / heat given off (1) White precipitate formed (1) which redissolves in excess (1) Green precipitate formed (1) With excess the green precipitate remains (1) One mark each for any two explanations from Acid is neutralised (1) Green precipitate is iron(II) hydroxide (1) White precipitate of zinc hydroxide (1) which will redissolve in excess (1)	4
B10(c)(i)	Oxidation number of iron increases / oxidation number of iron becomes more positive / iron(II) ions lose electrons (1)	1
B10(c)(ii)	Use (filter paper dipped into acidified) potassium manganate(VII) (1) Purple colour changes to colourless (1)	2
	Total:	10

(f) Describe the chemical test for sulfur dioxide.

test

observation

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Question	Answer	Marks
8(e)	(aqueous) ammonia (1) iron(II) sulfate – green ppt AND copper(II) sulfate blue ppt that redissolves in excess to give dark blue solution with NH ₃ (1) OR (aqueous) sodium hydroxide (1) iron(II) sulfate – green ppt AND copper(II) sulfate blue ppt (1)	2

Question	Answer	Marks
9(a)	acid rain / one effect of acid rain e.g. acidification of lakes or killing trees etc. (1)	1
9(b)	$4\text{FeS}_2 + 11\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 + 8\text{SO}_2$ correct formulae for reactants and products (1) balanced – dependent on correct formulae for reactants and products (1)	2
9(c)	food preservation / antioxidant / making sulfuric acid / bleaching wood pulp (1)	1
9(d)	movement – particles move faster (1) arrangement – particles spread out / particles no longer touching each other (1)	2
9(e)	has a simple (molecular) structure (1) weak intermolecular forces / weak attraction between molecules (1)	2
9(f)	test – (filter paper dipped in) acidified potassium manganate(VII) (1) observation goes from purple to colourless (1)	2

(c) Describe the chemical test for chlorine gas.

test

observation

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Question	Answer	Marks
1(a)	hydrogen (1)	1
1(b)	sulfur dioxide (1)	1
1(c)	ethane (1)	1
1(d)	methane (1)	1
1(e)	ozone (1)	1
1(f)	carbon monoxide (1)	1

Question	Answer	Marks
2(a)	all have 7 electrons in their outer shell (1)	1
2(b)(i)	(colourless) to brown (solution) (1)	1
2(b)(ii)	iodide (ion) loses electrons (1)	1
2(b)(iii)	chlorine (molecule) gains electrons (1)	1
2(c)	test – (moist red or blue) litmus paper (1) observation – bleached / goes white (1)	2
2(d)(i)	rate increases particles are more crowded / more particles in the same volume / increased number of particles per unit volume (1) more collisions per second / increased collision frequency (1)	2
2(d)(ii)	alternative reaction pathway (1) has lower activation energy (1)	2

(iii) Describe a test for chlorine.

test

result

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Question	Answer	Marks
A1(a)(i)	Iron / Fe (1)	1
A1(a)(ii)	Oxygen / O ₂ (1)	1
A1(a)(iii)	Hydrogen / H ₂ (1)	1
A1(a)(iv)	Aluminium / Al (1)	1
A1(a)(v)	Aluminium / Al (1)	1
A1(b)	⁴¹ K electrons 19 (1) neutrons 22 (1) ¹⁷ O ²⁻ electrons 10 (1) neutrons 9 (1)	4

Question	Answer	Marks
A2(a)	Arrangement: ordered / lattice / regular / layers / uniformly arranged / repeated pattern (1) Type of force: electrostatic (1)	2
A2(b)	Crystals: ions cannot move (1) Aqueous: ions can move (1)	2
A2(c)	Magnesium ion 2,8 (1) Chloride ion 2,8,8 (1)	2
A2(d)(i)	Hydrogen is <u>lower</u> in the reactivity series (than sodium) / sodium more reactive (than hydrogen) / hydrogen ions are reduced more easily (than sodium ions) (1)	1
A2(d)(ii)	2H ⁺ + 2e ⁻ → H ₂ (1)	1
A2(d)(iii)	Litmus paper / named indicator paper (1) Bleaches / loses its colour (1)	2

(b) Describe a chemical test for hydrogen.

test

observation if hydrogen present

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Question	Answer	Marks
3(a)(i)	A activation (energy) / E_a (1) B enthalpy (change) / ΔH (1)	2
3(a)(ii)	product level below reactant level	1
3(b)	'pops' with a lighted splint	1
3(c)	(moles of hydrogen =) $366 / 24000$ OR 0.01525 (1) (moles of Al =) 0.01525×0.67 OR 0.01017 (1) mass of Al = 0.27 (1)	3

Question	Answer	Marks
4(a)(i)	Any two from: M1 random (arrangement) (1) M2 (molecules are spread) far apart (1) M3 (molecules are) moving (very) fast (1) M4 random (motion)	2
4(a)(ii)	weak intermolecular forces	1
4(b)(i)	high melting point / high boiling point (1) conducts electricity in (aqueous) solution (1)	2
4(b)(ii)	$\text{Li} \rightarrow \text{Li}^+ + \text{e}^-$	1
4(b)(iii)	$\text{Br}_2 + 2\text{e}^- \rightarrow 2\text{Br}^-$	1

(d) Describe the test for oxygen.

test

observation

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Question	Answer	Marks
1(a)	sulfur dioxide	1
1(b)	iron(II) oxide	1
1(c)	aluminium oxide	1
1(d)	calcium oxide	1
1(e)	sulfur dioxide	1

Question	Answer	Marks
2(a)	21(%)	1
2(b)	carbon dioxide is an acidic oxide (1) reacts with base / reacts with alkali / neutralised by sodium hydroxide (1)	2
2(c)	air liquefied (1) <u>fractional</u> distillation (1)	2
2(d)	glowing splint (1) relights (1)	2
2(e)(i)	photochemical / redox	1
2(e)(ii)	lightning	1
2(f)	(more) skin cancer / (more) sunburn / (more) harm to eyes	1

(iii) Describe a chemical test for iron(III) ions.

test

observation

..... [2]

[Total: 10]

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Question	Answer	Marks
6(d)(ii)	anode – impure copper (1) cathode – (pure) copper (1)	2
6(d)(iii)	anode: $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$ (1) cathode: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ (1)	2

Question	Answer	Marks
7(a)	rate of forward reaction = rate of backward reaction (1)	1
7(b)	(composition) does not change (1) same number of moles of <u>gas</u> on left hand side and right hand side (1)	2
7(c)(i)	exothermic / enthalpy change is negative (1)	1
7(c)(ii)	rate increases because particles have more energy / particles moving faster / more successful collisions / more energetic collisions (1)	1
7(d)(i)	water (1)	1
7(d)(ii)	$\text{Fe}_3\text{O}_4 + 4\text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{FeSO}_4 + 4\text{H}_2\text{O}$ correct formulae (1) balanced equation – dependent on correct formulae (1)	2
7(d)(iii)	test – add (aqueous) sodium hydroxide / (aqueous) ammonia (1) observation – brown ppt / brown solid (1)	2

(iii) Describe a test for iron(III) ions.

test

observation

Page 4	Mark Scheme	Syllabus	Paper
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Question	Answer	Mark
A4(a)	peroxodisulfate: increases rate / doubling concentration doubles the rate (1) iodide: increases rate / doubling concentration doubles the rate (1)	2
A4(b)	lower the activation energy (1)	1
A4(c)(i)	oxidation number of iron decreases / they gain electrons / iron(III) gets reduced (1)	1
A4(c)(ii)	yellow $\rightarrow$ brown (1)	1
A4(c)(iii)	add (aqueous) sodium hydroxide / add (aqueous) ammonia (1) red-brown precipitate (1)	2
A4(d)	$2\text{Fe}^{2+} + \text{S}_2\text{O}_8^{2-} \rightarrow 2\text{Fe}^{3+} + 2\text{SO}_4^{2-}$ (1)	1
	Total:	8

Question	Answer	Mark
A5(a)	39% (2) If 2 marks not scored, molar mass of $\text{KClO}_3 = 122.5$ scores 1 mark	2
A5(b)	$\text{KClO}_3 = \frac{12.25}{122.5}$ OR 0.10 (mol) (1) moles $\text{O}_2 = 1.5 \times 0.01 = 0.15$ moles / idea of multiplying moles by 1.5 (1) volume of $\text{O}_2 = 3.6 \text{ dm}^3 / 3600 \text{ cm}^3$, correct unit must be included (1)	3

- (i) Describe a test for iron(II) ions.

test

observations

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Question	Answer	Marks
2(e)(i)	$\text{mol C} = \frac{14.4}{12} \quad \text{mol Cl} = \frac{21.3}{35.5} \quad \text{mol H} = \frac{0.600}{1}$ OR mol C = 1.20 mol Cl = 0.60 mol H = 0.60 (1) empirical formula is C₂HCl (1)	2
2(e)(ii)	C ₆ H ₃ Cl ₃	1

Question	Answer	Marks
3(a)	any three from: • diffusion • particles move (from place to place) / particles collide • random (movement) of particles / particles go anywhere / particles (move) in all directions / particles disperse • intermingling of particles / mixing of particle • (bulk movement of ink particles) from higher to lower concentration	3
3(b)(i)	(aqueous) sodium hydroxide / (aqueous) ammonia (1) (light) green precipitate (1)	2
3(b)(ii)	Fe ²⁺ → Fe ³⁺ + e ⁻	1
3(b)(iii)	lowers the activation energy	1
3(c)(i)	starch	1
3(c)(ii)	reflux / heat / boil (1) with (concentrated) hydrochloric acid (1)	2

- (d) Describe a chemical test to distinguish between aqueous iron(II) ions and aqueous iron(III) ions.

test

observations with aqueous iron(II) ions

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Question	Answer	Marks
5(a)	magnesium loses electrons and is oxidation (1) iron(II) ions gain electrons and is reduction (1)	2
5(b)	(some of the) electrons move (throughout the structure)	1
5(c)	magnesium is too reactive / magnesium is very reactive / magnesium is high in the reactivity series	1
5(d)	add (aqueous) sodium hydroxide / (aqueous) ammonia (1) iron(II) ions give green precipitate (1) iron(III) ions give red-brown precipitate (1)	3
5(e)(i)	the ions can move	1
5(e)(ii)	<i>anode</i> : chlorine / Cl ₂ (1) <i>cathode</i> : magnesium / Mg (1)	2
5(f)	does not corrode in water / does not corrode in air / unreactive	1

- (e) Describe a chemical test that can be used to distinguish between aqueous solutions of iron(II) sulfate and iron(III) sulfate.

chemical test

result with iron(II) sulfate

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Question	Answer	Marks
9(e)	(aqueous) ammonia / (aqueous) sodium hydroxide (1) iron(II) sulfate – green ppt AND iron(III) sulfate – rusty / red-brown / orange / orange-brown ppt (1)	2

Question	Answer	Marks
10(a)	a mixture (1)	1
10(b)	aviation fuel / heating (in the home) / cooking (stoves) (1)	1
10(c)(i)	alkane is $\text{C}_n\text{H}_{2n+2}$ which fits C_3H_8 if $n = 3$ (1) alkene is C_nH_{2n} which fits C_3H_6 if $n = 3$ (1)	2
10(c)(ii)	substitution (1) $\text{C}_3\text{H}_6\text{Cl}_2$ / $\text{C}_3\text{H}_5\text{Cl}_3$, etc. (1)	2
10(c)(iii)	orange to colourless / decolourises (1)	1
10(d)(i)	Any one from: $\begin{array}{c} \text{H}_3\text{C} \\ \\ \text{C}=\text{CH}_2 \\ \\ \text{H}_3\text{C} \end{array}$ $\begin{array}{c} \text{H} & \text{H} \\ & \diagdown \quad \diagup \\ & \text{C}=\text{C} \\ & \diagup \quad \diagdown \\ \text{H}_3\text{C} & \text{CH}_3 \end{array}$ $\begin{array}{c} \text{H}_3\text{C} & \text{H} \\ & \diagdown \quad \diagup \\ & \text{C}=\text{C} \\ & \diagup \quad \diagdown \\ \text{H} & \text{CH}_3 \end{array}$ $\begin{array}{c} \text{H} & \text{H} \\ & \diagdown \quad \diagup \\ \text{H}_3\text{C}-\text{C} & -\text{C} \\ & \\ \text{H} & \text{CH}_2 \end{array}$	1

(b) Describe a test for copper(II) ions.

test

observations

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Question	Answer	Marks
3(a)	(some of the) electrons can move / (some of the) electrons are mobile	1
3(b)	(aqueous) sodium hydroxide (1) (light) blue precipitate (insoluble in excess) (1) OR (aqueous) ammonia (1) (light) blue precipitate soluble in excess / dark blue solution formed in excess ammonia (1)	2
3(c)(i)	positive electrode: bubbles / effervescence / fizzes (1) negative electrode: turns pink (1) electrolyte: blue colour (of solution) fades (1)	3
3(c)(ii)	inert / (relatively) unreactive	1
3(d)	$\text{Cu}^{2+}(\text{aq}) + \text{Mg}(\text{s}) \rightarrow \text{Cu}(\text{s}) + \text{Mg}^{2+}(\text{aq})$ correct formulae and balancing (1) correct state symbols – dependent on correct formulae (1)	2
3(e)	mass of copper = 2.00 g (1) $\text{mol copper} = \frac{2.00}{64}$ and $\text{mol oxygen} = \frac{0.25}{16}$ OR $\text{mol copper} = 0.0313$ and $\text{mol oxygen} = 0.0156$ (1) empirical formula is Cu ₂ O (1)	3

- a few drops of aqueous ammonia are added to an aqueous solution containing copper(II) ions

.....

- excess aqueous ammonia is added to an aqueous solution containing copper(II) ions.

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Question	Answer	Marks
5(d)(i)	(light) blue precipitate (1) (dissolves to form) dark blue solution (1)	2
5(d)(ii)	+1	1
5(e)	Any three from: titrate (aqueous ammonia with hydrochloric acid) (1) repeat titration without indicator (1) evaporate filtrate to crystallisation point / heat until saturated solution formed (1) leave / cool and filter or leave / cool and dry (1)	3

Question	Answer	Marks
6(a)(i)	there are no mobile electrons	1
6(a)(ii)	high melting point	1
6(b)	weak intermolecular forces	1
6(c)	one pair of electrons in each overlap area AND two lone pairs on each sulfur atom AND 6 unpaired electrons on each chlorine atom	1

- (c) Aqueous ammonia is added dropwise until in excess to a small volume of aqueous copper(II) sulfate.

Describe the observations during this addition.

.....
..... [2]

- (d) Describe the observations when aqueous barium nitrate is added to aqueous copper(II) sulfate.

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Question	Answer	Marks
8(a)	C ₁₄ H ₃₀	1
8(b)	tetradecane	1
8(c)(i)	M1 fractional distillation / use of a fractionating column (1) M2 petroleum is heated / petroleum is vapourised / petroleum is boiled (1) M3 separation because of different boiling points (1)	3
8(c)(ii)	heating / cooking	1
8(c)(iii)	(chemical) feedstock	1

Question	Answer	Marks									
9(a)	substance which contains no water	1									
9(b)	blue	1									
9(c)	(pale) blue precipitate (1) soluble in excess resulting in dark blue solution (1)	2									
9(d)	white precipitate	1									
9(e)	<table border="1"> <tr> <td>electrodes</td><td>carbon</td><td>copper</td></tr> <tr> <td>anode</td><td>fizzing / bubbling (1)</td><td>gets smaller (1)</td></tr> <tr> <td>cathode</td><td>pink solid (1)</td><td>gets bigger / pink solid (1)</td></tr> </table>	electrodes	carbon	copper	anode	fizzing / bubbling (1)	gets smaller (1)	cathode	pink solid (1)	gets bigger / pink solid (1)	4
electrodes	carbon	copper									
anode	fizzing / bubbling (1)	gets smaller (1)									
cathode	pink solid (1)	gets bigger / pink solid (1)									
9(f)	ions can move in aqueous / ions cannot move in solid	1									

(e) To a sample of Cu(NO₃)₂(aq), a student adds aqueous ammonia drop by drop until it is in excess.

(i) Describe what is observed.

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Question	Answer	Total
B7(a)	Blue solution / brown gas / gas bubbles / metal disappears	1
B7(b)(i)	Copper(II) nitrate	1
B7(b)(ii)	Copper loses electron(s)	1
B7(c)	Moles of acid = 0.025×16 OR 0.4 (1) Moles of NO ₂ = 0.2 (1) Volume of NO ₂ = $4.8 \text{ dm}^3 / 4800 \text{ cm}^3$ (1)	3
B7(d)	$2\text{Cu}(\text{NO}_3)_2 \rightarrow 2\text{CuO} + 4\text{NO}_2 + \text{O}_2$	1
B7(e)(i)	Blue precipitate / blue solid (1) In excess becomes a dark blue solution (1)	2
B7(e)(ii)	Blue precipitate / blue solid (which does not redissolve)	1

(b) Aqueous ammonia is added to aqueous zinc ions.

Describe the observations when:

a few drops of aqueous ammonia are added

.....

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Question	Answer	Marks
4(a)	to improve crop yield / to improve plant growth / to add minerals lost when plants are harvested	1
4(b)	ammonia is produced (1) (ammonia) escapes (from the soil) / ammonia is a gas (1)	2
4(c)	relative molecular mass of ammonium sulfate = 132 (1) $\frac{28}{132} \times 100$ OR 21% / 21.212% (1) 21.2 (1)	3
4(d)	$2\text{NH}_3 + \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$ correct formulae (1) correct balance (1)	2
4(e)	reactants on the left and products on the right and reactant line below product line (1) arrow vertically upwards between reactants and products with ΔH or enthalpy change label (1)	2

Question	Answer	Marks
5(a)	oxidation because zinc loses electrons (1) reduction because copper(II) ions gain electrons (1)	2
5(b)	<i>few drops</i> : white precipitate (1) <i>excess</i> : precipitate dissolves / colourless solution (1)	2

- (c) Aqueous ammonia is added, with mixing, to a solution containing zinc ions until no further change occurs.

What observations would be made during this test?

.....

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Question	Answer	Marks
1(a)(i)	C	1
1(a)(ii)	D	1
1(a)(iii)	B	1
1(a)(iv)	A	1
1(a)(v)	D	1
1(b)(i)	20	1
1(b)(ii)	molecule containing two atoms	1

Question	Answer	Marks
2(a)(i)	zinc loses electrons which is oxidation (1) nickel ions gain electrons which is reduction (1)	2
2(a)(ii)	$\text{Zn} + \text{Ni}^{2+} \rightarrow \text{Zn}^{2+} + \text{Ni}$	1
2(b)	workable arrangement with two electrodes dipping in liquid connected to power supply with two wires and no obvious gaps in the wiring (1) nickel or the fork is negative electrode and silver is positive electrode (1) electrolyte is labelled silver ions OR soluble silver salt OR electrolyte (1)	3
2(c)	(at first) white precipitate (1) dissolves (in excess ammonia) / soluble in excess (ammonia) / colourless solution (in excess ammonia) (1)	2

- (b) Aqueous ammonia is added dropwise until in excess to a small volume of aqueous zinc chloride.

Describe the observations during this addition.

.....

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Question	Answer	Marks
8(f)(i)	bacterial oxidation / reaction with (acidified) potassium manganate(VII)	1
8(f)(ii)	sodium <u>ethanoate</u> and water	1

Question	Answer	Marks									
9(a)	(chemically) combined with water	1									
9(b)	white precipitate (1) soluble in excess resulting in colourless solution (1)	2									
9(c)	white precipitate	1									
9(d)	<table><tr><td>concentration</td><td>dilute</td><td>concentrated</td></tr><tr><td>anode</td><td>oxygen (and water) / O_2 (and H_2O) (1)</td><td>chlorine / Cl_2 (1)</td></tr><tr><td>cathode</td><td>hydrogen / H_2 (1)</td><td>hydrogen / H_2 (1)</td></tr></table>	concentration	dilute	concentrated	anode	oxygen (and water) / O_2 (and H_2O) (1)	chlorine / Cl_2 (1)	cathode	hydrogen / H_2 (1)	hydrogen / H_2 (1)	4
concentration	dilute	concentrated									
anode	oxygen (and water) / O_2 (and H_2O) (1)	chlorine / Cl_2 (1)									
cathode	hydrogen / H_2 (1)	hydrogen / H_2 (1)									
9(e)	ions can move in aqueous / ions cannot move in solid	1									

(b) Describe a chemical test for aluminium ions.

.....

.....

.....

..... [2]

(c) The aqueous aluminium sulfate formed is crystallised to make hydrated aluminium sulfate,

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Question	Answer	Marks
6(a)	$2\text{Al(s)} + 6\text{H}^+(\text{aq}) \rightarrow 2\text{Al}^{3+}(\text{aq}) + 3\text{H}_2(\text{g})$ balancing (1) correct state symbol dependent on formulae (1)	2
6(b)	add (aqueous) sodium hydroxide (1) white ppt that redissolves / white ppt. soluble in excess (1) OR add (aqueous) ammonia (1) white precipitate (1)	2
6(c)	M_r for aluminium sulfate is 342 (1) $x = 18$ (1)	2
6(d)	mixture of a metal with another element	1

Question	Answer	Marks
7(a)	global warming / greenhouse effect / climate change	1
7(b)	acid rain / damage to buildings / killing living things	1
7(c)(i)	to increase the surface area (1) more exposed particles / more collisions per second (1)	2
7(c)(ii)	particles move slower / particles have less kinetic energy (1) fewer successful collisions / fewer collisions above that of activation energy (1)	2
7(d)(i)	(sulfur or sulfur dioxide) gains oxygen	1

(c) A sample of sodium chloride is tested using a flame test.

State the colour of the flame seen in this test.

..... [1]

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Question	Answer	Marks
2(b)(iii)	Any three from: (potassium) floats on water / moves across the surface of water (1) (potassium) melts (1) bubbles / fizzes / effervesces (1) forms a colourless solution / piece gets smaller and smaller (1) lilac flame (1)	3
2(c)	yellow / orange (1)	1

Question	Answer	Marks
3(a)(i)	(moles of zinc oxide =) $3.50 / 81$ OR 0.0432 (1) (moles of HCl =) 0.0600 (1) idea that need 0.0864 moles of HCl / only 0.0300 moles of ZnO react (1)	3
3(a)(ii)	so all the acid is used up (1)	1
3(a)(iii)	filtration (1)	1
3(b)	barium nitrate or barium chloride AND any soluble sulfate (1)	1
3(c)(i)	nitric acid and sodium hydroxide (1)	1
3(c)(ii)	titration (1)	1

(ii) an atom that forms an ion that gives a lilac colour in a flame test

..... [1]

(iii) an atom of a monatomic gas

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Question	Answer	Marks
1(a)(i)	G	1
1(a)(ii)	H	1
1(a)(iii)	F	1
1(a)(iv)	E	1
1(a)(v)	D	1
1(b)	(number of protons) 24 (1) (number of neutrons) 29 (1)	2

Question	Answer	Marks
2(a)(i)	oxidation and reduction occur simultaneously	1
2(a)(ii)	CO removes oxygen from iron oxide / CO gains oxygen from the iron oxide	1

(b) gives a yellow flame test colour

..... [1]

(c) dissolves to form a dark purple aqueous solution

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Annotation	Meaning
	point has been noted, but no credit has been given or blank page seen
	error in number of significant figures
	pages are linked together
	used to highlight part of the response
	key point attempted / working towards marking point / incomplete answer / response seen but not credited / blank page seen
	unclear response / query

(c) Describe the observations when aqueous silver nitrate is added to aqueous zinc chloride.

..... [1]

(d) Dilute and concentrated aqueous zinc chloride are electrolysed separately using carbon

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Question	Answer	Marks
8(f)(i)	bacterial oxidation / reaction with (acidified) potassium manganate(VII)	1
8(f)(ii)	sodium <u>ethanoate</u> and water	1

Question	Answer	Marks									
9(a)	(chemically) combined with water	1									
9(b)	white precipitate (1) soluble in excess resulting in colourless solution (1)	2									
9(c)	white precipitate	1									
9(d)	<table><tr><td>concentration</td><td>dilute</td><td>concentrated</td></tr><tr><td>anode</td><td>oxygen (and water) / O₂ (and H₂O) (1)</td><td>chlorine / Cl₂ (1)</td></tr><tr><td>cathode</td><td>hydrogen / H₂ (1)</td><td>hydrogen / H₂ (1)</td></tr></table>	concentration	dilute	concentrated	anode	oxygen (and water) / O₂ (and H₂O) (1)	chlorine / Cl₂ (1)	cathode	hydrogen / H₂ (1)	hydrogen / H₂ (1)	4
concentration	dilute	concentrated									
anode	oxygen (and water) / O₂ (and H₂O) (1)	chlorine / Cl₂ (1)									
cathode	hydrogen / H₂ (1)	hydrogen / H₂ (1)									
9(e)	ions can move in aqueous / ions cannot move in solid	1									

2 You are provided with solutions **R** and **S**.

(a) (i) Do the tests on **R** shown in the table.

Record your observations in the table.

test no.	test	observations
1	To 1 cm depth of R in a test-tube, add aqueous sodium hydroxide drop-by-drop until a change is seen. Add excess aqueous sodium hydroxide.	
2	To 1 cm depth of R in a test-tube, add aqueous ammonia drop-by-drop until a change is seen. Add excess aqueous ammonia.	
3	To 1 cm depth of R in a test-tube, add 3 drops of dilute nitric acid and then add 1 cm depth of aqueous silver nitrate.	

[6]

Conclusions

(ii) Name the cation in **R**.

cation

[1]

(iii) Name the anion in **R**.

anion

[1]

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Question	Answer	Marks
General points R is chromium(III) chloride S is iron(II) sulfate For gases: to gain credit for the name of the gas produced, the test must be at least partially correct. Solutions: colourless is not equivalent to clear and clear is not equivalent to colourless No credit is given for conclusions based upon incorrect observations.		
2(a)(i)	(test 1) (grey-)green ppt (1) dissolves (in excess) (1) green solution (1) (test 2) (grey-)green ppt (1) insoluble (in excess) (1) (test 3) white ppt (1)	6
2(a)(ii)	(cation is) chromium(III)	1
2(a)(iii)	(anion is) chloride	1

(b) (i) Do the tests on **S** shown in the table.

Record your observations in the table.

Test and name any gases produced.

Describe the test and the positive result for any gases you named.

test no.	test	observations
1	To 1 cm depth of S in a test-tube, add aqueous sodium hydroxide drop-by-drop until a change is seen. Add excess aqueous sodium hydroxide.	
2	To 1 cm depth of S in a boiling tube, add 1 cm depth of aqueous hydrogen peroxide drop-by-drop. Keep the solution for use in test 3.	
3	To 1 cm depth of the solution from test 2 in a test-tube, add aqueous sodium hydroxide drop-by-drop until a change is seen. Add excess aqueous sodium hydroxide.	
4	To 1 cm depth of S in a test-tube, add 3 drops of dilute nitric acid and then add 1 cm depth of aqueous barium nitrate.	

Conclusions

(ii) Name the cation responsible for the observations in test 1.

cation [1]

(iii) Name the cation responsible for the observations in test 3.

cation [1]

(iv) Name the anion in **S**.

anion [1]

[Total: 20]

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Question	Answer	Marks
2(b)(i)	(test 1) green ppt (1) insoluble (in excess) (1) (test 2) brown solution (1) effervescence (1) glowing splint relights (1) oxygen (1) (test 3) red-brown ppt (1) insoluble (in excess) (1) (test 4) white ppt (1)	9
2(b)(ii)	(cation is) iron(II) (1)	1
2(b)(iii)	(cation is) iron(III) (1)	1
2(b)(iii)	(anion is) sulfate (1)	1

2 You are provided with solution **W** and solid **X**.

Do the following tests recording all of your observations at each stage.

Tests on solution W

- (a) Put 1 cm depth of **W** into a boiling tube. Add 1 cm depth of aqueous sodium hydroxide. Gently warm the mixture.

Test the gas given off.

Describe the test and its result.

Identify the gas.

Keep the mixture for use in (b).

test for gas and result

.....

identity of gas

[3]

- (b) Add 2 cm depth of dilute nitric acid to the mixture from (a).

Then add 1 cm depth of aqueous barium nitrate.

Record your observations.

.....

..... [1]

- (c) Identify the cation and the anion in **W**.

cation anion

[2]

5070/31

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Question	Answer	Marks
2(a)	M1 damp red litmus M2 blue M3 ammonia / NH ₃	3
2(b)	white ppt	1
2(c)	M1 ammonium M2 sulfate	2
2(d)	M1 blue solution formed M2 effervescence M3 limewater becomes cloudy M4 carbon dioxide / CO ₂	4
2(e)	M1 blue ppt M2 insoluble in excess	2
2(f)	M1 blue ppt M2 dissolves in excess M3 dark blue solution	3
2(g)	M1 copper(II) M2 carbonate	2

Tests on solid X

- (d) Put one of the samples of **X** into a boiling tube and use a measuring cylinder to add 7 cm³ of dilute nitric acid.

Record your observations.

Test the gas given off.

Describe the test and its result.

Identify the gas.

Keep the mixture for use in (e) and (f).

observations

.....

test for gas and result

.....

identity of gas

.....

[4]

- (e) Use a measuring cylinder to transfer 2 cm³ of the mixture from (d) into a test-tube.

- (g) Identify the cation and the anion in **X**.

cation anion [2]

[Total: 17]

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Cambridge O Level – Mark Scheme
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Question	Answer	Marks
2(a)	M1 damp red litmus M2 blue M3 ammonia / NH ₃	3
2(b)	white ppt	1
2(c)	M1 ammonium M2 sulfate	2
2(d)	M1 blue solution formed M2 effervescence M3 limewater becomes cloudy M4 carbon dioxide / CO ₂	4
2(e)	M1 blue ppt M2 insoluble in excess	2
2(f)	M1 blue ppt M2 dissolves in excess M3 dark blue solution	3
2(g)	M1 copper(II) M2 carbonate	2

- (a) Do a flame test on solution **W**. Describe the method you use.

method

.....

.....

.....

observations

.....

conclusion

.....

[4]

- (b) (i) To 1 cm depth of **W** in a test-tube, add 1 cm depth of aqueous chlorine. Keep this mixture

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Question	Answer	Marks
2(a)	M1 dips / soaks (clean) rod / splint / wire in W M2 places in blue / roaring / colourless / non-luminous / air-hole open flame (and observes initial colour) OR M1 puts solution W in a spray bottle. M2 spray solution into blue / roaring / colourless / non-luminous / air-hole open flame (and observe initial colour) M3 lilac / purple M4 potassium ions present	2
2(b)(i)	M1 red / brown solution formed / black solid formed M2 may contain iodide / displaces iodine	2
2(b)(ii)	blue-black colour	1
2(c)	M1 add nitric acid / hydrochloric acid and barium nitrate / chloride to W M2 colourless solution / no (white) precipitate formed	2
2(d)	M1 (pale) yellow precipitate M2 iodide present	2
2(e)	red/brown solution formed / black solid formed	1
2(f)(i)	M1 white precipitate M2 soluble in excess sodium hydroxide / Colourless solution with excess sodium hydroxide M3 contains zinc ions or aluminium ions	3

- (a) Do a flame test on **one** sample of **A**. Describe the method you use.

method

.....

.....

observations

.....

conclusion

.....

[4]

- (b) (i) Place the other sample of **A** in a test-tube and add 3 cm depth of dilute hydrochloric acid.

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Question	Answer	Marks
2(a)	M1 dip wire / splint / rod into solid M2 put wire / splint into blue / roaring / airhole open / non-luminous / colourless flame M3 blue – Green colour in flame. M4 copper ions / Cu^{2+} present.	4
2(b)(i)	M1 effervescence / fizzing / bubbling. M2 (bas turns) limewater milky / cloudy. M3 carbon dioxide / CO_2 formed. M4 blue-green solution (formed). M5 carbonate (ions) / CO_3^{2-} present	5
2(b)(ii)	no solid remains undissolved / effervescence stops	1
2(c)(i)	M1 blue precipitate M2 (forms) deep blue solution (in excess)	2
2(c)(ii)	blue precipitate (insoluble in excess)	1
2(d)	copper carbonate, CuCO_3	1
2(e)	M1 chloride white precipitate and bromide cream / off white precipitate and iodide (pale) yellow precipitate. M2 result for B off white / cream precipitate M3 bromide / Br^- ions present	3

Test the gas given off.

Describe the test and its result.

Identify the gas.

Keep the mixture for use in **(b)** and **(c)**.

(d) Identify the cation **and** the anion in **Y**.

cation anion

[2]

— — —

5070/32

Cambridge O Level – Mark Scheme
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Question	Answer	Marks
1(a)	M1 full set of data provided i.e. all initial and final temperatures recorded in table M2 temperature readings recorded to 0.5 °C M3 correct temperature rises recorded i.e. all subtractions correct M4 initial increase in temperature M5 indication of levelling off or decreasing M6–8 temperature rises (corrected for errors in M3) within 1.0 °C of supervisor's results: 5 or 6 (3) 3 or 4 (2) 1 or 2 (1)	8
1(b)	M1 points plotted correctly M2 one straight line of best fit drawn through first 4 points and one straight line of best fit drawn through last 3 points M3 intersection of both lines	3
1(c)	volume of Q at the point of intersection of the two lines within 0.25 cm ³	1
1(d)	M1 moles NaOH = (25 × 1.25 / 1000 =) 0.03125 (mol) M2 concentration H ₂ SO ₄ = M1 × 1000 / 1c × 2	2
1(e)	M1 gradient less steep M2 more energy transferred (from solution to surroundings)	2
1(f)	improves the precision	1

Question	Answer	Marks
2(a)	M1 colourless solution formed M2 effervescence M3 limewater becomes milky / cloudy M4 carbon dioxide / CO ₂	4
2(b)	M1 white ppt M2 dissolves in excess M3 colourless solution (is formed)	3
2(c)	M1 white ppt M2 dissolves in excess	2